1 Network meta-analysis (NMA) extensions

We now extend our previously defined ordinal regression models for diagnostic test accuracy meta-analysis to accommodate the network meta-analysis framework, allowing for the inclusion of studies that assess multiple index tests and enabling comparative analysis across tests. These extensions follow the arm-based approach of Nyaga et al., but tailored to our ordinal regression framework.

1.1 Background: The Nyaga et al. Arm-Based NMA Model for Diagnostic Test Accuracy

Before presenting our network meta-analysis (NMA) extensions of the ordinal regression DTA models, we first review the arm-based NMA approach for diagnostic test accuracy as proposed by Nyaga et al.

Consider a setting with T tests and S studies, where not all tests are evaluated in each study. For a certain study s, let $(Y_{s,t}^{[1]}, Y_{s,t}^{[0]})$ denote the true positives and true negatives, $(N_{s,t}^{[1]}, N_{s,t}^{[0]})$ the diseased and healthy individuals, and $(\pi_{s,t}^{[1]}, \pi_{s,t}^{[0]})$ the "unobserved" sensitivity and specificity respectively with test t in study s. The study serves as a block where all diagnostic accuracy tests are hypothetically evaluated, though some may be missing.

Given study-specific sensitivity and specificity, the distribution of true positives and true negatives among the diseased and the healthy individuals follows:

$$Y_{s,t}^{[d]} \mid \pi_{s,t}^{[d]} \sim \text{Binomial}(\pi_{s,t}^{[d]}, N_{s,t}^{[d]}),\ s = \{1, \ldots, S\},\ t = \{1, \ldots, T\}. \tag{1.1}$$

where $d = 1$ refers to the diseased individuals and $d = 0$ to the healthy individuals.

The arm-based single-factor design with repeated measures model is written as follows:

$$\pi_{s,t}^{[d]} = \Phi\left(\mu_t^{[d]} + \eta_s^{[d]} + \delta_{s,t}^{[d]}\right) \tag{1.2}$$

$$\begin{pmatrix} \eta_s^{[0]} \\ \eta_s^{[1]} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \boldsymbol{\Sigma}\right) \tag{1.3}$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} \tag{1.4}$$

$$(\delta_{s,1}^{[d]}, \delta_{s,1}^{[d]}, \ldots \delta_{s,T}^{[d]}) \sim N(\mathbf{0},\ \text{diag}((\tau^{[d]})^2)) \tag{1.5}$$

where: $\mu_t^{[1]}$ and $\mu_t^{[0]}$ are the mean sensitivity and specificity in a hypothetical study with random-effects equal to zero respectively. $\eta_s^{[d]}$ is the study effect for healthy individuals ($d = 1$) or diseased individuals ($d = 2$) and represents the deviation of a particular study s from the mean sensitivity ($d = 1$) or specificity ($d = 0$), inducing between-study correlation. The study effects are assumed to be a random sample from a population of such effects.

The between-study variability of sensitivity and specificity and the correlation thereof is captured by the parameters σ_1^2, σ_2^2, and ρ respectively. $\delta_{s,t}^{[d]}$ is the error associated with the sensitivity ($d = 1$) or specificity ($d = 2$) of test t in the sth study. Conditional on study s, the repeated measurements are independent with variance constant across studies such that: $(\tau^{[d]})^2 = ((\tau_1^{[d]})^2, \ldots, (\tau_T^{[d]})^2)$ is a T dimensional vector of homogeneous variances.

In essence, the model separates the variation in the studies into two components: the within-study variation $\text{diag}((\tau^{[d]})^2)$, referring to the variation in the repeated sampling of the study results if they were replicated, and the between-study variation $\boldsymbol{\Sigma}$, referring to variation in the studies' true underlying effects.

1.2 NMA Extension of the Jones Model

We extend the Jones et al. model to the network meta-analysis framework, incorporating multiple tests while maintaining the Box-Cox transformation approach for handling continuous thresholds.

For each test t in study s, the test accuracy measures at the kth cutpoint are represented as:

$$\begin{aligned}
\text{Sp}_{s,t,k} &= \Phi\left(\frac{-(C_{t,k} - \beta_{s,t}^{[0]})}{\exp\left(\gamma_{s,t}^{[0]}\right)}\right) \\
\text{Se}_{s,t,k} &= 1 - \Phi\left(\frac{-(C_{t,k} - \beta_{s,t}^{[1]})}{\exp\left(\gamma_{s,t}^{[1]}\right)}\right)
\end{aligned} \tag{1.6}$$

Where the cutpoints $C_{t,k}$ for test t are obtained through either log or Box-Cox transformation of the continuous threshold values:

$$C_{t,k} = \begin{cases} \log(\text{thr}_{t,k}) & \text{if box_cox} = 0 \\ \frac{\text{thr}_{t,k}^{\lambda_t} - 1}{\lambda_t} & \text{if box_cox} = 1 \text{ and } \lambda_t \neq 0 \\ \log(\text{thr}_{t,k}) & \text{if box_cox} = 1 \text{ and } \lambda_t = 0 \end{cases} \tag{1.7}$$

1.2.1 Hierarchical Structure

Following the Nyaga et al. framework, we decompose the location and scale parameters using a two-level hierarchical structure:

$$\begin{aligned}
\beta_{s,t}^{[d]} &= \beta_{\bullet,t}^{[d]} + \eta_{\beta,s}^{[d]} + \delta_{\beta,s,t}^{[d]} \\
\gamma_{s,t}^{[d]} &= \gamma_{\bullet,t}^{[d]} + \eta_{\gamma,s}^{[d]} + \delta_{\gamma,s,t}^{[d]}
\end{aligned} \tag{1.8}$$

Where:

- $\beta_{\bullet,t}^{[d]}$ and $\gamma_{\bullet,t}^{[d]}$ are the mean location and log-scale parameters for test t in disease group d
- $\eta_{\beta,s}^{[d]}$ and $\eta_{\gamma,s}^{[d]}$ are study-level random effects shared across all tests within study s
- $\delta_{\beta,s,t}^{[d]}$ and $\delta_{\gamma,s,t}^{[d]}$ are test-specific deviations for test t in study s

1.2.2 Between-Study Model

The study-level random effects (shared across tests) follow a bivariate normal distribution:

$$\begin{pmatrix} \eta_{\beta,s}^{[0]} \\ \eta_{\beta,s}^{[1]} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \boldsymbol{\Sigma}_\beta\right) \tag{1.9}$$

$$\begin{pmatrix} \eta_{\gamma,s}^{[0]} \\ \eta_{\gamma,s}^{[1]} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \boldsymbol{\Sigma}_\gamma\right) \tag{1.10}$$

Where:

$$\boldsymbol{\Sigma}_\beta = \begin{bmatrix} (\sigma_\beta^{[0]})^2 & \rho_\beta \sigma_\beta^{[0]} \sigma_\beta^{[1]} \\ \rho_\beta \sigma_\beta^{[0]} \sigma_\beta^{[1]} & (\sigma_\beta^{[1]})^2 \end{bmatrix} \tag{1.11}$$

$$\boldsymbol{\Sigma}_\gamma = \begin{bmatrix} (\sigma_\gamma^{[0]})^2 & \rho_\gamma \sigma_\gamma^{[0]} \sigma_\gamma^{[1]} \\ \rho_\gamma \sigma_\gamma^{[0]} \sigma_\gamma^{[1]} & (\sigma_\gamma^{[1]})^2 \end{bmatrix} \tag{1.12}$$

The test-specific deviations are modeled independently:

$$\begin{aligned}
\delta_{\beta,s,t}^{[d]} &\sim \text{Normal}(0, \tau_{\beta,t}^{[d]}) \\
\delta_{\gamma,s,t}^{[d]} &\sim \text{Normal}(0, \tau_{\gamma,t}^{[d]})
\end{aligned} \tag{1.13}$$

1.2.3 Incorporating Covariates

The model accommodates study-level covariates through the mean parameters. For test t:

$$\begin{aligned}
\beta_{\bullet,t}^{[0]} &= \mathbf{X}_{nd,t}\boldsymbol{\beta}_{\bullet,t}^{[0]} \\
\beta_{\bullet,t}^{[1]} &= \mathbf{X}_{d,t}\boldsymbol{\beta}_{\bullet,t}^{[1]} \\
\gamma_{\bullet,t}^{[0]} &= \mathbf{X}_{nd,t}\boldsymbol{\gamma}_{\bullet,t}^{[0]} \\
\gamma_{\bullet,t}^{[1]} &= \mathbf{X}_{d,t}\boldsymbol{\gamma}_{\bullet,t}^{[1]}
\end{aligned} \tag{1.14}$$

Where $\mathbf{X}_{nd,t}$ and $\mathbf{X}_{d,t}$ are the design matrices for covariates in the non-diseased and diseased groups for test t, and $\boldsymbol{\beta}_{\bullet,t}^{[d]}$ and $\boldsymbol{\gamma}_{\bullet,t}^{[d]}$ are the corresponding coefficient vectors.

1.2.4 Scale Parameter Transformation

The scale parameters are obtained through transformation of the raw scale parameters:

$$\sigma_{s,t}^{[d]} = \begin{cases} \log(1+\exp(\gamma_{s,t}^{[d]})) & \text{if softplus} = 1 \\ \exp(\gamma_{s,t}^{[d]}) & \text{if softplus} = 0 \end{cases} \tag{1.15}$$

Where $\gamma_{s,t}^{[d]} = \gamma_{\bullet,t}^{[d]} + \eta_{\gamma,s}^{[d]} + \delta_{\gamma,s,t}^{[d]}$.

1.2.5 Summary Estimates

The summary accuracy measures for test t at threshold k are obtained by evaluating at the mean parameter values with baseline covariate values:

$$\begin{aligned}
\text{Se}_{\bullet,t,k} &= 1 - \Phi\left(\frac{-(C_{t,k} - \beta_{\bullet,t}^{[1],\text{baseline}})}{\sigma_{\bullet,t}^{[1],\text{baseline}}}\right) \\
\text{Sp}_{\bullet,t,k} &= \Phi\left(\frac{-(C_{t,k} - \beta_{\bullet,t}^{[0],\text{baseline}})}{\sigma_{\bullet,t}^{[0],\text{baseline}}}\right)
\end{aligned} \tag{1.16}$$

Where:

$$\begin{aligned}
\beta_{\bullet,t}^{[d],\text{baseline}} &= \mathbf{x}_{\text{baseline}}^T \boldsymbol{\beta}_{\bullet,t}^{[d]} \\
\gamma_{\bullet,t}^{[d],\text{baseline}} &= \mathbf{x}_{\text{baseline}}^T \boldsymbol{\gamma}_{\bullet,t}^{[d]} \\
\sigma_{\bullet,t}^{[d],\text{baseline}} &= \begin{cases} \log(1+\exp(\gamma_{\bullet,t}^{[d],\text{baseline}})) & \text{if softplus} = 1 \\ \exp(\gamma_{\bullet,t}^{[d],\text{baseline}}) & \text{if softplus} = 0 \end{cases}
\end{aligned} \tag{1.17}$$

1.2.6 Variance Decomposition

Following Nyaga et al., the total between-study variance for each test can be decomposed as:

$$\text{Var}[\beta_{s,t}^{[d]}] = (\sigma_{\beta}^{[d]})^2 + (\tau_{\beta,t}^{[d]})^2 \tag{1.18}$$

$$\text{Var}[\gamma_{s,t}^{[d]}] = (\sigma_{\gamma}^{[d]})^2 + (\tau_{\gamma,t}^{[d]})^2 \tag{1.19}$$

The proportion of variance explained by test-specific heterogeneity is:

$$\text{Prop}_{\beta,t}^{[d]} = \frac{(\tau_{\beta,t}^{[d]})^2}{(\sigma_{\beta}^{[d]})^2 + (\tau_{\beta,t}^{[d]})^2} \tag{1.20}$$

$$\text{Prop}_{\gamma,t}^{[d]} = \frac{(\tau_{\gamma,t}^{[d]})^2}{(\sigma_{\gamma}^{[d]})^2 + (\tau_{\gamma,t}^{[d]})^2} \tag{1.21}$$

This decomposition allows assessment of how much heterogeneity is due to general study effects versus test-specific variations.